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MILITARY SPECIFICATION

# PRIMER DETONATOR M16A1 LOADING, ASSEMBLING, AND PACKING

*This specification has been approved by the Department of Defense for the use of the Departments of the Army, the Navy, and the Air Force.*

## 1. SCOPE

1.1 This specification covers the following primer detonators:

4 to 5 seconds delay.

8 to 15 seconds delay.

## 2. APPLICABLE DOCUMENTS

2.1 The following specifications, drawings, and publications, and all drawings and specifications referenced thereon, of the issue in effect on date of invitation for bids form a part of this specification:

### SPECIFICATIONS

MILITARY

MIL-C-13739 — Composition, Delay

MIL-G-2550 — General Specification For Ammunition Except Small Arms Ammunition

### DRAWINGS

ORDNANCE CORPS

73-8-83 — Primer Detonator, M16A1, 8 to 15 Seconds Delay Assembly.

73-8-84 — Primer Detonator, M16A1, 4 to 5 Seconds Delay Assembly

81-3-65 — Chronograph Assembly

### PUBLICATIONS

ORD-SIP-M16-101—Primer Detonators, M14 and M16A1 Loaded.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 **Material.**—Material and parts shall be in accordance with applicable specifications and drawings. (See 6.1.)

3.2 The primer detonator shall comply with all requirements specified on Drawings 73-8-83, or 73-8-84, as applicable, and with all requirements specified in applicable specifications.

**3.3 Delay powder and ignition powder.**—The moisture content of the delay powder and ignition powder at the loading operation shall not exceed the amount specified in Specification MIL-C-13739 when determined as specified in 4.4.

**3.4 Static functioning.**

**3.4.1** *Functioning time, 4- to 5-second delay.*—The 4- to 5-second primer detonators (less detonator holder assemblies if equipment for testing with detonator holder assemblies is not available) shall have static functioning

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times from 3.50 to 5.00 seconds inclusive, when tested as specified in 4.5. The average functioning time shall be not less than 4.10 seconds nor more than 4.40 seconds.

3.4.2 *Functioning time, 8- to 15-second delay.*—The 8- to 15-second primer detonators (less detonator holder assemblies if equipment for testing with detonator holder assemblies is not available) shall have static functioning times from 7.75 to 12.50 seconds inclusive when tested as specified in 4.5. The average functioning time shall be not less than 9.7 seconds nor more than 10.5 seconds.

3.4.3 *Effective functioning.* — The primer detonator shall cause the relay charge to function effectively when tested as specified in 4.5.

**3.5 Proving ground requirement for the primer detonators.**

3.5.1 *Primer detonators that are issued as integral parts of fuzes.*—Primer detonators that are manufactured to be issued as integral parts of fuzes shall be tested in accordance with the loading specification for the fuzes in which they are used.

3.5.2 *Primer detonators that are made to be issued separately.* — Separately issued primer detonators shall be tested in accordance with 4.6. All primer detonators when assembled to an M112A1 or M113A1 fuze shall cause satisfactory detonation of the bomb when fired.

**3.6 Workmanship.**—All parts shall be free from chips, dirt, grease, rust, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Lot.**—A lot shall consist of all primer detonators loaded by one manufacturer in one unchanged process, in accordance with the same drawing, the same drawing revision, same specification and the same specification revision date. Each lot shall contain:—

(a) Primers of one interfix lot number from one manufacturer.
(b) Detonators of one interfix lot number from one manufacturer.
(c) Relays of one interfix lot number from one manufacturer.
(d) Primer-detonators of one interfix lot number from one manufacturer.
(e) Delay powder and ignition powder from one lot each.
(f) Metal parts of one interfix lot number from one manufacturer.

**4.2 Sampling.** — Number of samples, acceptance and rejection criteria used for determination of lot acceptance for tests as specified in 4.4 and 4.5 shall be in accordance with Standard Inspection Procedure ORD–SIP–M16–101. Ten sample primer detonators shall be selected from each lot for proving ground test as specified in 4.6.

**4.3 Inspection.** — Inspection shall be as specified in Specification MIL–G–2550.

**4.4 Determination of moisture content of delay powder and ignition powder.**—The allowable moisture content and the method of determining the percentage thereof shall be in accordance with Specification MIL–C–13739 to determine compliance with 3.3.

**4.5 Static functioning test of primer-detonators.** — Sample primer detonators (less detonator holder assemblies if equipment for testing with detonator holder assemblies is not available) shall be selected from each lot and tested at a temperature of 70° ± 10° F. in a fixture simulating the firing of the primer detonator in the M112A1 or M113A1 Tail Bomb Fuze. The delay shall be measured to determine compliance with 3.4, by means of chronograph conforming to Ord-



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nance Corps Drawing 81–3–65, or a chronograph satisfactory to the inspector.

**4.6 Proving ground tests for primer detonators.**—Ten samples shall be selected from each lot of primer detonators that are made to be issued separately from fuzes. They shall be assembled to five M112A1 and five M113A1 fuzes from accepted lots and subjected to the following tests:

4.6.1 *4- to 5-second delay primer detonators.* — Five samples in 100-lb and five in 500-lb H.E. bombs shall be dropped on water at a speed of not less than 250 mph. The altitudes of release shall be approximately 100 feet for the 100-lb bombs and, approximately 50 feet for the 500-lb bombs. Observation shall be made for detonation of the bomb to determine compliance with 3.5.2. The delay time of functioning after impact shall be recorded for information only.

4.6.2 *8- to 15-second delay primer detonators.*—Five samples in 100-lb and five in 500-lb H.E. bombs shall be dropped, three of each on land and two of each on water, at a speed of not less than 250 mph. The altitude of release shall be approximately 100 feet on land and approximately 50 feet on water. Observation shall be made for detonation of the bomb to determine compliance with 3.5.2 The delay time of functioning after impact shall be recorded for information only.

4.6.3 *Acceptance.*—A lot will be considered acceptable provided that all primer detonators comply with the requirements of 3.5. If 9 of the primer detonators cause satisfactory detonation of the bomb the lot will be considered eligible for retest under the provisions of 4.6.4.

4.6.4 *Retest.*—A retest as provided for in 4.6.3 may be made, if requested by the contractor, using 20 primer detonators for the retest. The lot shall be considered acceptable provided the 20 primer detonators cause satisfactory detonation of the bomb.

## 5. PREPARATION FOR DELIVERY

**5.1 Packaging, packing, labeling and marking.** — Packaging, packing, labeling, and marking shall be as specified on the drawing.

**5.2 Data cards.** — Data card information shall be as specified in Specification MIL–G–2550.

## 6. NOTES

**6.1** It has been found advisable that the following conditions be observed during loading operations:

Loading room condition. — A temperature of approximately 75°F. and a maximum relative humidity of 40 percent shall be maintained. Metal components for loading shall be brought into the loading room only after having been subjected in a vented oven to a minimum temperature of 100°F. for at least 8 hours. Loading of these components shall be accomplished after they have reached the temperature of the air-conditioned loading room.

**6.2 Ordering data.** — Procurement documents should specify the title, number and date of this specification.

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